

Vinh Ly

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Summary

Software Engineer with **18 years** of experience architecting and shipping distributed systems at scale. Led Search Relevance infrastructure at YouTube serving billions of daily queries with sub-100ms latency targets. Built the Checkout service at Whatnot handling **2M+ monthly transactions** with 99.99% uptime. Scaled Google's Search and Knowledge Graph from 3B to 15B daily queries, mentoring 15+ engineers across three teams. Expert in **Node.js**, **Python**, **Kubernetes**, and **PostgreSQL** with deep background in API design, observability, and reliability engineering. Seeks to drive end-to-end system ownership and technical mentorship in a high-ownership engineering organization.

Core Skills

Languages and Frameworks: JavaScript, HTML, CSS

Backend: Node.js, NestJS, Python, TypeScript, PHP, REST APIs, GraphQL

Infrastructure: AWS, Kubernetes, Docker, Terraform, GCP, CI/CD

Data: PostgreSQL, MongoDB, Redis, Elasticsearch, BigQuery, NoSQL

Authentication: JWT, OAuth 2.0, OIDC, Auth0, SSO, JWKS

Experience

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Leading Search Relevance backend serving billions of daily queries, architecting ranking and retrieval pipelines with sub-100ms p99 latency requirements across the team.
- Owning technical roadmap for the Search platform, partnering with ML engineers to ship relevance improvements to production with measurable engagement lift.
- Establishing observability standards and on-call rotations, reducing alert fatigue by 40% through automated runbooks and SLO-based alerting.
- Mentoring 5 engineers on distributed systems design, code review practices, and career growth within a fast-moving product organization.
- Driving adoption of **Kubernetes** and **CI/CD** modernization across the Search infrastructure, improving deployment frequency by 3x.

Staff Software Engineer, Whatnot

05/2024 - 02/2025 | Remote

- Designed and shipped the Checkout service powering **\$100M+** in monthly gross merchandise value, integrating **Stripe** and Braintree with **Node.js** microservices.
- Refactored Checkout from a monolithic PHP service to **NestJS** microservices, cutting p99 latency by **50%** and reducing infrastructure costs by **\$200K** annually.
- Maintained 99.99% uptime across payment processing services during peak events, handling **10x normal traffic** spikes without degradation.
- Implemented **JWT** authentication with Auth0 and JWKS validation, securing all API endpoints with OIDC-compliant flows.
- Established engineering standards for **schema design** in **PostgreSQL** and MongoDB, reducing query times by **30%** across core services.
- On call for payment and order processing systems, triaging incidents and shipping fixes within SLA windows for critical revenue-impacting services.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Scaled Knowledge Graph backend serving **3B to 15B** daily queries with 99.99% availability, leading a team of 6 engineers on end-to-end system ownership.
- Reduced Knowledge Panels serving latency from **500ms to 50ms** p99 through query optimization, caching layers, and **PostgreSQL** index restructuring.
- Shipped entity indexing pipeline processing **10B documents** daily using **Python** and distributed **Kubernetes** workloads on **GCP**.
- Mentored 15+ engineers across Search and Assistant teams, conducting performance reviews and driving promotions for 8 senior engineers.
- Partnered with ML and product teams to ship multimodal search ranking improvements, increasing click-through rates by **12%** in A/B tests.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Built core ranking pipeline for Image Search serving 5B+ daily requests, developing **Python** data processing jobs on **GCP** infrastructure.
- Designed indexing schema for document retrieval system handling **10B+ documents** with **BigQuery** analytics for quality monitoring.
- Shipped ranking features for web search with measurable improvements in relevance metrics across **microservices** architecture.
- Reduced on-call incident resolution time by **25%** through improved **observability** dashboards and automated alerting in production systems.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Shipped key features for **Exchange Server 2007** Web Access, building server-side validation and **PHP** frontend components for enterprise email.
- Developed **PostgreSQL** database schemas and stored procedures for the Outlook protection system, optimizing query performance by **40%**.
- Participated in on-call rotations for production mail services, resolving service-impacting incidents within SLA windows.
- Collaborated with product managers to translate requirements into technical specifications for calendar and contact synchronization features.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Built internal tooling for test case management using **C#** and **.NET** frameworks, reducing manual QA overhead by **15%** for the Exchange team.
- Automated regression test suites for the Outlook Web Access service, improving test coverage and accelerating release cycles for the team.

Education

B.S. Honors

Virginia Tech | 2004 - 2007